

Solar Panel Light Tracker

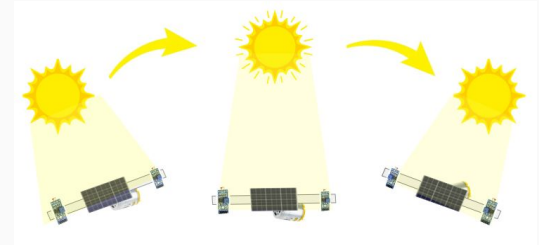
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Introduction

Our project involves building a prototype solar panel that will move towards the position of the maximum sunlight.

The importance of this project: This project aims to make solar panels more efficient, which in turn allows us to create more energy from renewable sources (such as solar energy)



Companies such as Array Technologies have come up with solutions in which many motors tilt the solar panel towards the sun

Introduction

Limitations with other solutions:

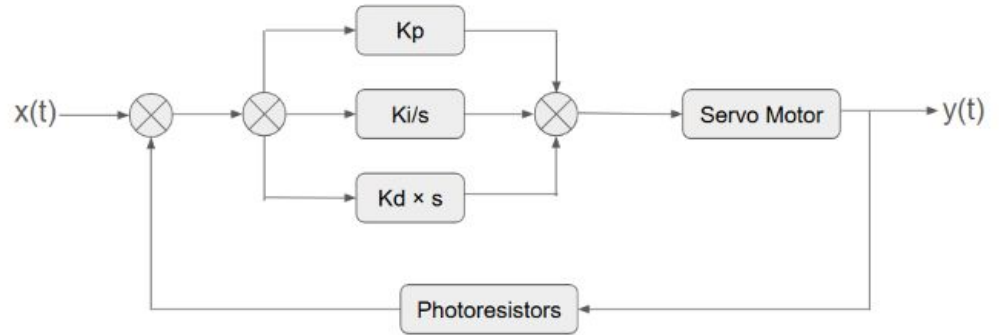
1. Cost ineffective or expensive
2. Doesn't have a scope with individual panels an in residential areas

Our project plans to create an inexpensive and lightweight solution



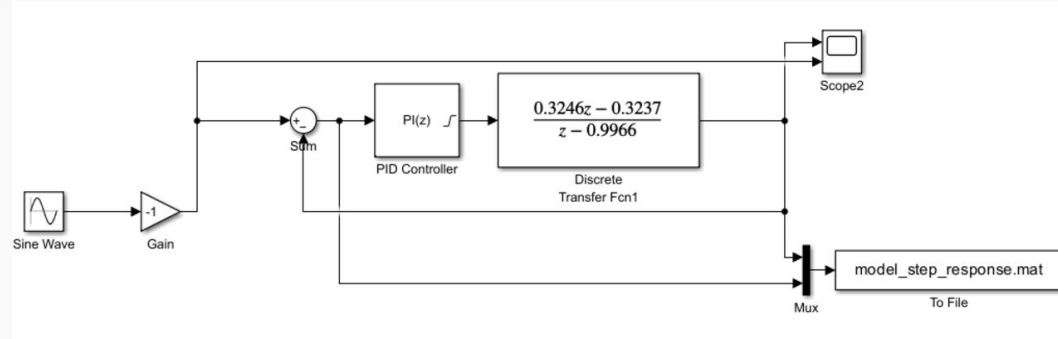
System Architecture

The system will use a PID (Proportional Integral Derivative) control. The plant/process of our system is a servo motor. The difference between the photoresistor will act as the feedback to our system.



System Architecture

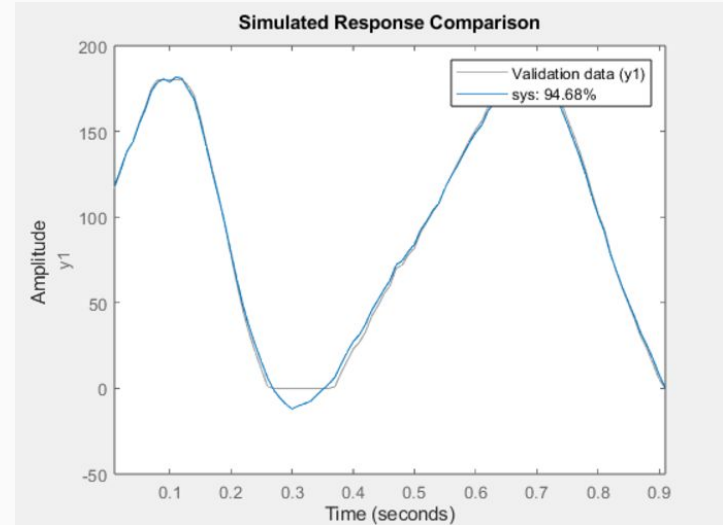
The continuous transfer function was determined using the system identification toolbox. This continuous transfer function was converted to a discrete transfer function using the c2d function. The discrete transfer function is displayed in the figure on the right and the transfer function is below.



$$Transfer Function = \frac{0.3246s + 0.08954}{s + 0.344}$$

System Evaluation

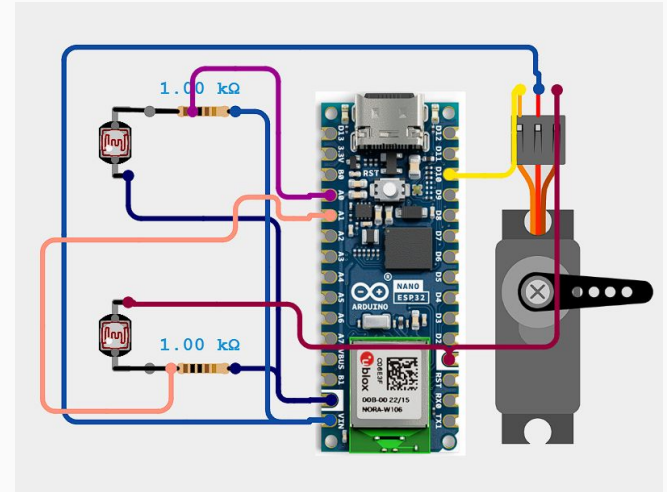
This demonstrates the final response for the system compared to the predicted response. It shows that the actual motor reflects the simulation with 94.68% accuracy!



Hardware System

Our system will consist of two photoresistors at the opposite end of the solar panel. The position of the sun will determine the readings on the photoresistors.

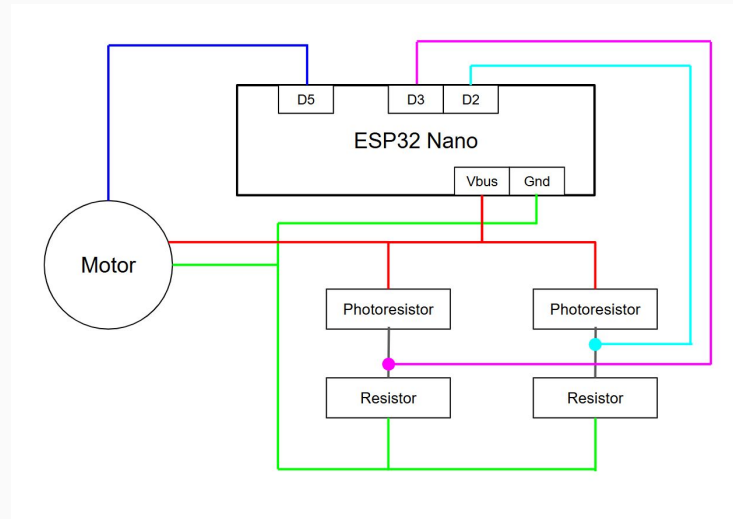
Two pins on the arduino, D2 and D3, will read the values of the photoresistors. These values will be used to move the motor.



Hardware System

The arduino is programmed to send an output through pin D5 to the motor. This output will be determined by the inputs read from D2 and D3.

Note that the motor must be connected to the arduino's power supply, V_{bus} (5V), and ground pin.

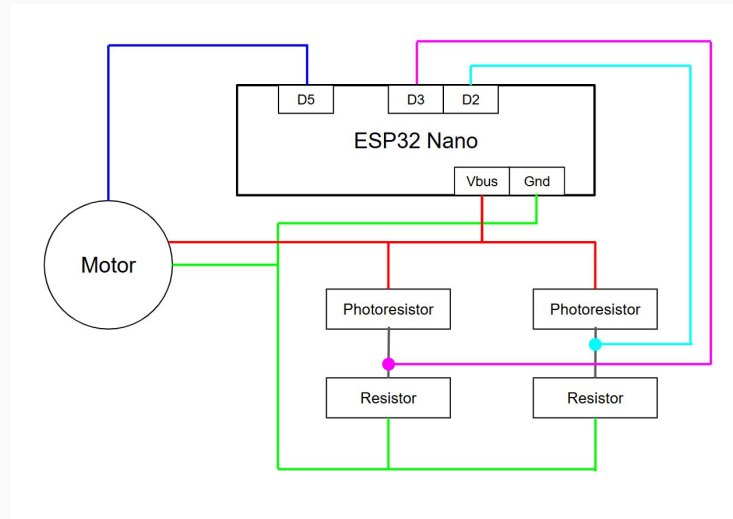


Hardware System

In case of the photoresistors, the arduino's power supply, V_{bus} (5V) must be connected to the photoresistors.

These photoresistors must be connected to the resistors, which are connected to ground.

The input pins, D2 and D3, should be connected in between the photoresistor and the resistor.



Software

- The FreeRTOS Arduino code was used as a template for the code used in this projects.
- The FreeRTOS Arduino code was modified for readability and servo controls were added.
- Our code uses the combined voltage of the photoresistors as the input to the PID controller. The output of the PID controller is used as the input for the servo motor
- The direction of the servo motor movement is determined by the difference in voltage between the photoresistors

Budget

Component	Price
PhotoResistors	\$5
ESP-32	\$27
Wires	Free
Resistors	Free
Servo	Free
StyroFoam	Free
Total	\$32

Testing and Results



Lessons Learned

- This project taught us how PID systems can be implemented in a variety of systems and increase efficiency in a wide range of applications.
- We gained experience using MATLAB and the Arduino IDE
- We learned how to implement sensors, such as photoresistors, in a complex system